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## **M 305 DUCTING, DAMPER AND BY PASS STACK**

### **1 SCOPE OF WORK**

The Bidder shall supply the following in accordance with Part 4, subsection 1.2, “Summary of Work”. A complete system of exhaust ducting, dampers and exhaust stack for the Combined Cycle Plant which shall include but not be limited to the exhaust ducting, expansion joints, vibration isolators, supporting steelwork and flanges, blanking plates, inspection doors, silencing equipment, control and damper, bypass stack(existing), instrumentation and control equipment, insulation and cladding and any other plant and equipment required to complete the system, and the required electrical equipment specified in Clause 8 herein.

### **2 CODES AND STANDARDS**

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards and equivalent.

The Codes and Standards of the following organizations extracted from Part 4, General Technical Requirements are specifically applicable to the design manufacture and testing of equipment included in this specification:

|      |                                                    |
|------|----------------------------------------------------|
| AISC | American Institute of Steel Construction           |
| AISI | American Iron and Steel Institute                  |
| ASME | American Society of Mechanical Engineers           |
| ASTM | American Society for Testing and Materials         |
| AWS  | American Welding Society                           |
| SSPC | Standard Specification for prime and Coat painting |
| IEC  | International Electrotechnical Commission          |
| IEEE | Institute of Electrical and Electronic Engineers   |
| JIS  | Japan Industrial Standard                          |

|      |                                            |
|------|--------------------------------------------|
| SPLN | Standard of Perusahaan Umum Listrik Negara |
|------|--------------------------------------------|

### **3 DESIGN AND OPERATING CONDITIONS**

The whole of the exhaust ducting system shall be designed to withstand a back pressure of not less than 50 mbar g.

The ducting shall be designed to reduce the accumulation of deposits to a minimum; to keep the resistance to the exhaust gas flow as low as practicable and to give the required distribution of gas flow at inlet to the HRSG. If necessary to ensure proper gas distribution, specially designed deflector plates capable of adjustment, shall be provided. Such deflectors shall be installed such that they can readily be repaired or replaced.

An adequate number of expansion joints shall be installed to accommodate duct expansion and prevent undue thermal stresses in the ductwork under all operating conditions so that the gas turbine exhaust flange and HRSG inlet flange can be anchored to prevent the onward transmission of any significant forces and moments to the gas turbine or HRSG.

Instrument points shall be provided in the ducts for monitoring, testing and checking purposes.

Gas tight opposite access doors of an approved standard type shall be provided to give access to ducts for inspection and maintenance. The Bidder shall provide to minimum two (2) sets for HRSG inlet duct and one (1) set for HRSG outlet duct of gas tight opposite access doors. They shall be of the flanged type and not less than 600 mm x 600 mm size. The ducts shall be adequately stiffened to support the doors. Any door or hatch shall be capable of being securely locked and interlocks shall be provided to prevent any attempted Gas Turbine start up with the doors open.

Attention shall be paid to the provision of landing platforms local to access doors for gas ducting. To ensure safety and convenience of access, landing platforms shall be provided inside the ducts where there is no obvious landing within 1 meter of the door itself. Such platforms are to be easily replaceable.

The design of the exhaust ducting system shall be such that under all operating modes and site conditions, there shall not be any condensation within the system.

The HRSG ducting shall be designed so that each economizer inlet feed water temperature can be operated under the ambient temperature during the combined cycle start up.

Any exposed surface shall be sufficiently lagged and clad for personnel protection end to maintain metal temperature above the dew point of the exhaust gas. Preformed lagging shall be secured to the ductwork and clad with suitable sheeting to reduce heat by radiation to a minimum and to protect against the entry of rain water. Lagging shall conform to the specifications as stated in the Technical Specification.

The ducting shall be designed so that rain water falling on the cladding shall run off.

The Bidder shall provide all necessary weatherproofing and sealing where ducts pass through the wall or roof of the building.

The exhaust duct system shall be designed to avoid drumming, vibration and any resonance with the exhaust gas pulsations under all operating conditions.

HRSG's steel structure and steel supports shall be designed to avoid vibration and shall meet the specific requirement of Part 4, Technical Specifications S-401 and S-402.

The exhaust ducting shall incorporate damper (gas isolating) for controlling and directing the flue gas to the bypass stack or to the HRSG.

The damper shall also be designed and constructed to minimize heat losses through the bypass stack and enable inspection and maintenance of the HRSG to be undertaken while the gas turbine is in operation.

The mechanical design shall ensure that distortions due to rapid thermal cycling are reduced to a minimum and that sealing efficiency is maintained at a high level after extended service.

The control of the damper shall be synchronized to enable the HRSG start-up heating rate to be regulated without recourse to restricting the gas turbine operation in any way.

The damper shall be arranged for automatic remote control in addition to local manual operation.

The damper shall be located so as to avoid creating pockets from which unburn gases cannot be easily purged before turbine ignition.

Damper shall be proven design and approved by the owner.

The Blanking plate shall be located between the control damper and HRSG inlet and installed when HRSG is maintained. The plate design, including lowering, sealing and support mechanism shall be acceptable to the owner.

## **4 DETAILED REQUIREMENTS**

### **4.1 Ducts**

The ducts shall be of all welded steel plate construction suitably stiffened where necessary by means of angles, tees or flats secured to the outside of the ducts.

The duct wall and roof shall not be less than 9 mm thick and the duct floor not less than 10 mm thick including a corrosion allowance of 3 mm and shall be adequately supported.

The welding of ducts shall be continuous internally but may be discontinuous externally.

All joints between sections of ducts shall be made by means of angles of approved size welded to the plating. Joint bolts shall not be less than 15 mm diameter with a pitch of not more than 100mm. All bolted joints shall be made with sealing tape approximately 3 mm thick over the full width of the flange or with an approved type of jointing material.

The ductwork shall contain drainage points of adequate size for rain removal which shall be piped to the station drain system.

All ductwork surfaces shall be shot-blasted to remove all mill scale before applying the 1st coat of paint at shop or site.

The material of the ductwork, guide vanes, expansion joints and all other parts of the exhaust system shall be suitable in every respect for thermal cycling and continuous operation at the high temperature conditions pertaining to the peak load on the gas turbine at the highest ambient temperature. Stress levels shall be kept to levels which ensure a creep life at the maximum temperature of at least 100,000 hours.

Sliding joints shall not be used in the ductwork.

The Bidder shall quote the joints requested and listed in the “Schedule of Expansion Joints”. At design temperature, the expansion joints shall have an additional expansion absorption capability of 25% without damage.

A detailed description and calculations of the allowance for thermal expansion shall be given in Technical Schedule.

The expansion joints shall be fabric or metal design.

The expansion joints of fabric design shall have fabric bellows elements consisting of an outer layer of Fluoropolymer reinforced with fiberglass, needled or ceramic insulating mats with mesh and vapor barrier. The joint shall be protected internally against flue-gas and externally against rain water.

All expansion joints shall be flanged for removal without disturbing the main sections of the ducting. The Lagging shall be sectionalised for removal local to the expansion joints. Bolts for the flanged joints shall be not less than 15 mm diameter.

## **4.2 Damper**

The exhaust gas ducting system of gas turbine shall connect to gas flow diverter. The Damper shall be of single blade design. The damper shall comprise modulating control/quick acting isolation damper. The damper shall be capable to modulate the gas flow between bypass stack and HRSG port or to insulate either outlet entirely. The blade shall be capable of being held for extended periods in any end positions (HRSG close or bypass stack close position) on load change over from open to combine cycle without disturbing Gas Turbine load.

The damper blade assembly shall consist of a single pivoting assembly and be driven at toggle arm torque tube. The blade shall be designed to accommodate thermal growth at specified condition. A proven thermal compensation system shall be incorporated to prevent misalignment with the drive system.

The seals of the damper shall be fitted so that when it is closed and the seal air system is not operating, they are assisted by the differential pressure.

A locking device shall be provided for use when the damper is closed.

The shaft(s) and blade of the damper shall not reduce in the open position the nominal open cross-sectional area of the HRSG port.

The damper shall be a single of flap design with the shaft(s) offset to the edge(s) of the duct.

The entire operating gear shall be external to the duct and be capable of inspection and maintenance while the main plant is on line. All toggle link bearings shall be maintenance free and incorporate self aligning features.

The blade shall not be rigidly attached to the pivoting support structure. Allowance shall be made for differential expansion. The blade pivot tube shall be constructed of pipe in stainless steel.

The blade shall be of single skin construction of stainless steel and insulation shall be fitted to the nominal cold side of the blade to limit thermal distortion.

The damper shall be supplied with a sealing system providing, in the closed position, a sealing efficiency of 99.98%, or better, on duct cross- sectional area, without the assistance of seal air.

An air barrier shall be incorporated in the sealing system for both outlets to provide 100% isolation. This shall provide a pressure zone of at least 50 mm WC above the normal duct pressure and be connected through a differential pressure switch to an audible alarm. The pressure switch tapping point shall be located at the furthest point possible from the seal air intake. The air barrier fan's shut-off valve shall remain open in the event of a fan failure and the standby fan shall take over.

In the open position (to HRSG), the damper shall provide a minimum obstruction to the gas-flow.

Full details shall be provided of similar dampers supplied for equivalent applications, sizes and conditions.

The damper seal design shall minimize the fan power required to maintain a pressure higher than the exhaust gas pressure in the damper interspace. The seal air fans and the associated equipment shall be included under this Contract.

Seals shall be furnished in segments, not longer than 600 mm in length, adequate to compensate for differential expansion between the seal and its mounting. They shall be easily replaceable at site.

Heat transfer through and across the blade when closed to the bypass stack shall not exceed 3 kW/m<sup>2</sup> (with seal air) and 8 kW/m<sup>2</sup> (without seal air).

With internally insulated damper, frame mounted seal landing surfaces and/or seal mountings shall not be directly welded to the frame. Adequate allowances shall be made for the differences in thermal expansion.

The seals shall give a flexible margin seal of not less than 20 mm width in hot condition and bridge a gap between the blade and the frame. Seals which are compressed between the blade and the framework are not acceptable.

Sealing efficiencies shall be guaranteed for a minimum of five years. Details shall be provided of tests previously undertaken, to show that damper can achieve the required sealing efficiency.

The seals shall be wholly metallic and be made of materials which

- Maintain adequate mechanical properties for a minimum of five years
- Are not permeable
- Are capable of accommodating distortions in the damper framework and blade up to 10 mm per 1000 mm run.

The shaft bearings shall

- Have no requirement for lubrication,

- Be unaffected by particulates and dust build ups,
- Be unaffected by the maximum expected temperature, and
- Be self-aligning.

The shaft sealing system shall

- be of the fully metallic labyrinth type
- not rely on the elastic properties of its materials of construction
- not be deformed during differential expansion of the shaft
- contain no fibrous material
- not inhibit the self-aligning capabilities of the bearings

The operating gear for the damper shall be fitted with a graduated indicator and shall be designed, so that the damper position can be retained between full open and shut positions with the plant in operation.

Damper blade, spindles and bearings shall be of robust design of approved materials to continuously withstand the highest exhaust temperatures, pressures and vibration anticipated under service conditions without distortion, seizure or deterioration to ensure personnel and machinery safety.

The damper for turbine/HRSG set shall be operated by hydraulic rams driven from a common electric motor driven power pack. Manual operation of the damper shall be by hand pump.

Damper Hydraulic Rams shall have

- Self aligning bushes
- Standard cushioning
- Stainless steel rods
- Bleed nipples
- Viton seals

The pressure relieving and fast opening functions for the By-Pass damper shall be via a nitrogen gas pre-charged accumulator system in the power pack.

The power pack shall be contained in a weatherproof sheet steel enclosure and have but not limited to the followings:

- Ventilation
- The capability of being drained quickly and easily with a base forming an oil drain with one central drain plug at the front
- An oil reservoir, with a removable cover, incorporated as an integral part of the enclosure. The cover shall be marked “warning. This is an oil tank”
- Lifting lugs for loading, unloading and site installation
- Pressure and return line filtration
- All electrical connections fed to a terminal box, inside the enclosure, with klippon type terminals 1.5 mm<sup>2</sup> and 10% spare terminals
- Oil resistant cables
- ANSI 316 piping with plated carbon steel fittings
- Pressure test points on all services
- Output pressure gauges on all pumps
- Accumulator pressure gauges
- An oil tank level indicator
- A tank floats witch for low level alarm

The hydraulic system shall include, but not limited to

- A variable capacity main pump for normal operation and a separate variable capacity accumulator charging pump for emergency operation with mutual automatic back-up,
- An accumulator charging pump for pilot pressure
- Separate accumulators for emergency operation and pilot pressure
- Separate flow regulation for return line of gas-isolator flap
- A counterbalance valve in the piping for the by-pass gas-isolator (both directions) for speed and load control
- Pipe rapture protection (counterbalance type) on the hydraulic rams of the by-pass gas-isolator

- A service port relief valve for the in-stroke on the by-pass gas-isolator to limit pressure in pressure relief mode
- Provision for manual operation by hand pump of gas-isolator

Loadings imposed on the Gas-Isolators shall be kept to a minimum by use of suitably located expansion joints and support structures.

#### **4.3 Silencers (Not Applicable)**

Silencers of stainless steel construction shall be provided in the gas turbine ducting and Bypass stack to reduce the plant noise levels in accordance with the General Technical Requirements, Indonesia National Standard and OSHA including high, medium and low frequency.

The silencer acoustic panels shall be designed for a minimum operating life of that with the plant operating continuously at 100% maximum base load mass flow rating and with maximum gas temperatures. The silencer shall be removable from the ductwork without dismantling or removing any other ductwork and Bypass stack.

The infill and panels shall be fully resistant to the exhaust gas temperature and the worst anticipated combination of corrosive gas turbine combustion products. Precautions shall be taken to prevent settling or packing of the infill materials. The in fill material shall be vermin proof.

#### **4.4 Bypass Stack (Not Applicable)**

The Exhaust Bypass Stack shall be self supporting and shall terminate at a 45 m minimum height above ground level which will ensure that the ground pollutants are within the limits specified. The stack shall be designed and constructed to approved standard and in all respects to the approval of the Owner. The stack shall be cladded to provide an appearance matching that of the HRSG stack, and be thermally insulated where required to protect personnel. The cladding shall be painted with red and white bands to meet local air craft safety requirements.

The Bypass Stack shall be fitted with a drain and access manhole at its base. At the top, around the full circumference, a platform shall be provided complete with a safety rail and access ladder with safety hoops. The design of

the Bypass Blast Stack shall be such that under all operating modes and site conditions there must be no condensation within the stack.

The Bypass Stack shall be equipped with suitable earthing, lighting and other electrical pertinences as per appropriate system of the electrical specifications.

A dispersion study to determine the stack height shall be performed by the Bidder in order to meet the ambient air standard as mentioned in Part 4, Subsection 2.16, however the minimum height shall be 45 m above ground level.

The studies/calculations/analysis for wind oscillation effect on bypass stack shall be performed by the Bidder.

#### **4.5 Painting**

The painting for the exhaust ducting system shall meet the specific requirement of Part, Technical Specifications S-420 and S-430.

### **5 PERFORMANCE TESTING AND GUARANTEES**

#### **5.1 General**

The Bidder shall comply with the requirements of M-014 Inspection and Test.

The Owner shall have complete access to Bidder's or sub-Bidder's shops for reviewing progress of work and performing inspections.

The damper shall be shop assembled, tested, and shipped in the minimum number of pieces.

Bidder shall submit his standard shop testing details with the proposal.

The Owner shall be given sufficient notice of any shop tests to be performed on the assembled expansion joints. Test results shall be documented and evaluated to assure that test requirements and acceptance limits have been satisfied.

Air leakage tests shall be performed at Site for all ductwork in accordance with a procedure acceptable to the Owner. As a minimum the ductwork shall be pressurized to 500mm of water column and the pressure shall not decay by more than 5% of the initial pressure in 30 minutes.

Test “Acceptance Limits” will be submitted to the Owner for review and acceptance.

## **5.2 Specific Guarantees**

Bidder’s proposal shall, include a detailed description of the expected useful life of expansion joint for respective location in the duct system, under the conditions specified. Bidder shall fully guarantee all expansion joints for two years starting from plant startup. Any failed joints shall be repaired or replaced at Bidder’s expense during this period.

The damper shall be guaranteed to cause no more than 10 mm WG pressure loss when operating fully open at the specified conditions. The details of the expected open area and the method of calculating pressure drop across the damper shall be provided.

## **6 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)**

Quality Assurance/Quality Control shall conform to the requirements of Part 4 Subsection 2.7.

QA/QC procedures shall include, but not be limited to, the following:

- Welding
- Nondestructive Testing
- Heat Treating
- Packaging and Shipping
- Hydrostatic and Seat Leakage Testing

The quality control documentation shall include, but not limited to, the following:

- Mill Test Reports
- NDE test Reports
- Seat Leakage Test Reports
- All other Documentation Required by Applicable codes and standards

## **7 INSTRUMENTATION AND CONTROLS**

### **7.1 General**

The Bidder shall be responsible for the design, supply and installation of instrumentation and control devices listed and described in this technical specification and part 4, Technical Specification I-001, I-002 and I-100.

Bidder shall provide as a minimum the instrument connections on ducting, damper and bypass stack.

Bidder shall furnish all, the inline devices such as thermowells and thermocouples, test thermowells and flow elements etc.

### **7.2 Damper Drive and Controls**

Local control switches, modulating and open/close damper drives, pushbuttons indicating lights and motor starters shall be provided.

Damper operators shall be capable of a minimum of four consecutive open and close cycles against full differential pressure without damage to the electric motor driven power pack. Bidder shall state the time period required for four consecutive open and close cycles against full differential pressure.

The strength of the actuator mount shall be based on the actuator stall torque, and shall not exceed 10% of the Yield strength of any component many mode of stress.

The Bidder shall permanently identify with instrument or equipment numbers, all instruments, damper, motors and any other equipment as required by Owner in accordance with Part 4, Technical Specification I-001

### **7.3 Flue Gas Monitoring**

The Bidder shall comply with requirements of 7.10 Flue Gas Monitoring of M-301 HRSG.

## **8 ELECTRICAL SPECIFICATIONS**

### **8.1 General**

The Bidder shall furnish and install all electrical equipment and accessories required for the Damper and Expansion Joints. Those equipment shall include but not be limited to the followings:

- Motors
- All electrical control, monitoring and instrumentation devices and other appurtenances required for this Specification M-305 Ducting, Damper and Bypass Stack shall be provided by the Bidder.

All expansion joints carrying flue gas shall be equipped with two 120 mm<sup>2</sup> copper jumpers, one at the top and one at the bottom of the expansion joints. The jumpers shall be connected to the bottom of frames of the expansion joint and the duct work on side, for grounding continuity.

### **8.2 Motors**

For detailed electrical requirements, see Part 4 Technical Specification E-210 for Low Voltage Motors, E-211 for Medium Voltage Motors.

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